EXPERIMENT NO. 7

INVESTIGATION OF TURBULENT BOUNDARY LAYER GROWTH IN THE PRESENSE OF A STING MOUNT

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INVESTIGATION OF TURBULENT BOUNDARY LAYER GROWTH IN THE PRESENSE OF A STING MOUNT

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I. Introduction

Turbulent boundary layer theory is a very important topic in aerodynamics because almost all real flows over aircrafts and aerospace structures are affected by viscosity. In an ideal and inviscid flow, the fluid can slip along a surface without losing momentum due to friction. However, in real flow, the no-slip condition requires the fluid directly at the wall to have a velocity of zero. The velocity continues to increase as the distance away from the surface increases until it approaches the free stream velocity value. This phenomenon creates what is known as a velocity profile.

Although boundary layers are often small compared to the overall size of an aerodynamic body, they can have a huge effect on the flow in terms of frictional effects. Boundary layers contribute to skin friction drag, cause flow separation, and can change the area effected by the flow. For wind tunnels, boundary layers seen by testing mounts can affect the quality of flow seen by other testing bodies in the tunnel.

This current experiment is a part of a larger project of wind tunnel measurements in the Virginia Tech Stability Wind Tunnel. In earlier testing, the boundary layers in the empty test section were measured in the absence of any testing mount. Those tests provided a baseline database of boundary layer growth on the wind tunnel walls without any other body present. In this experiment, a sting mount was added to help determine how the support body would alter the flow and resulting frictional effects. This step is needed before adding a full testing model in the tunnel, because the sting itself can disturb the flow by creating blockage effects, pressure gradients, and chaotic vortex flow. Once the effects of the sting mount are better understood, future testing can use the sting to mount an airfoil or smaller aircraft models.

This phase of testing is particularly important for computational fluid dynamics, also known as CFD. CFD simulations are useful only if they accurately model and represent the testing mount, the test body, and the facility the tests are preformed in. If the wind tunnel boundary layers that occur on the walls, pressure gradients, and blockage effects are not modeled correctly, the calculated flow fields will not match the flows seen in real life. This provides the motivation to study the turbulent boundary layer growth on testing mounts, as they are needed to provide validation data for CFD models.

A major reason why the sting mount must be studied separately is that it can create blockage effects. Blockage occurs when an object inside a wind tunnel takes up space in the tunnel that forces the flow to move around the body. Since the same amount of air must pass through a smaller area, the flow around the sting mount must accelerate. This changes the velocities and pressure fields compared to what would occur in open air. In this specific experiment the sting does not only take up space within the tunnel, but it also changes the wall boundary layers by affecting the pressure distributions near the floor and ceiling. Similarly, the sting also creates an adverse pressure gradient. An adverse pressure gradient means that the pressure increases in the direction of the flow. This happens close to the front of the sting because the incoming flow is forced to slow down as it approaches the stagnation point. When the flow slows down, the static pressure rises. The outer flow has enough momentum to move through this pressure rise, but the flow with lower momentum doesn't, so it moves into a horseshoe vortex. This vortex can stretch around the sting, which brings fluid of higher momentum closer to the wall, which changes the measured boundary layer profile. Therefore, comparing the differences between the boundary layer growth with and without changes to the sting geometry can be used to determine the influence of the sting geometry.

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Also in this experiment, two different geometries of the sting were compared. First, a standard sting with no fillet at the bottom was used. The absence of a fillet provides a reference case for how the sting affects the wall boundary layer. In the second set of tests, a fillet was added to the bottom of the sting to reduce the horseshoe vortex effects seen in the first set of tests. A comparison can then be made to determine the effects of the fillet on the velocity profiles and boundary layer thicknesses.

The theoretical foundation for this experiment stems from turbulent boundary layer growth. The boundary layer thickness, δ, is defined as the distance from the wall to 0.99 of the local free stream value, U_∞. The thickness depends mostly on the streamwise distance, the flow speed U_∞, flow density ρ_∞, and viscosity μ_∞ through the Reynolds number:

$$Re_x = \frac{\rho_\infty U_\infty x}{\mu_\infty} \tag{1}$$

where x is the distance from the leading edge of the body where the boundary layer begins. For a laminar boundary layer on a flat plate, the boundary layer thickness can be approximated by:

$$\frac{\delta}{x} = \frac{5.0}{\sqrt{Re_x}} \tag{2}$$

However, turbulent boundary layers are more complicated because they contain more chaotic flow resulting in bigger velocity fluctuations. Because of this, there is not a simple exact equation for turbulent boundary layer thickness like there is for a laminar flat plate. Instead, scientists use a power law approximation based on past experimental trends. The turbulent relationship used in this experiment is

$$\frac{\delta}{x} = \frac{0.37}{{Re_x}^{1/5}} \tag{3}$$

This means that typically the boundary layer thickness is expected to increase downstream as x increases but decrease as the Reynolds number increases. Since Reynolds number includes both the distance and the free stream velocity, the approximate dependance can be written as

$$\delta \propto x^{4/5} {U_\infty}^{-1/5} \tag{4}$$

This relationship is useful as a comparison but won't perfectly match the experiment at hand. This wind tunnel is not an ideal flat plate, the effective boundary layer starting point is not known, and the sting mount creates a pressure gradient and blockage effects that are not represented in the power law model.

Additionally, the pressure measurements from the boundary layer rake were translated into velocity measurements via Bernoulli's equation. For steady and incompressible flow, Bernoulli's equation states that static pressure plus dynamic pressure is equal to the total pressure. The relationship can be shown as

$$p_0 = p + \frac{1}{2}\rho u^2 \tag{5}$$

where p_0 is the total pressure measured by one of the probes on the pressure rake, p is the static pressure at that location, ρ is the air density, and u is the flow velocity. Rearranging this equation for flow velocity gives:

$$u = \sqrt{\frac{2(p_0 - p)}{\rho}} \tag{6}$$

This equation allowed the pressure data from each rake probe to be converted into a velocity profile.

The aims of this study are:

1. To determine how the turbulent boundary layer thicknesses changed between the upstream and downstream pressure rake locations

2. To evaluate whether the measured boundary layer thickness followed the power law relationship, where increasing Reynolds number should generally reduce δ at a fixed location.
3. To use the upstream measurements to estimate the effective boundary layer starting point and determine how well the power law model predicted the downstream measurements.
4. To compare the two sting leading edge geometries by determining whether the filleted geometry produced a thicker or thinner downstream boundary layer than the no-fillet geometry.
5. To identify the experimental limitations that affected the reliability and accuracy of measured velocity profiles and boundary layer thicknesses.

These objectives were achieved by measuring boundary layer profiles that were perpendicular to walls using a pneumatic boundary layer rake. The rake was placed at two different streamwise locations in the test section, and data was collected over a specific sweep of Reynolds numbers from approximately 1.15 to 3.9 million per meter. At each condition, the rake measured stagnation pressures at 30 positions, while the local static pressure was measured at each panel. These pressure measurements were converted to velocity profiles using Bernoulli's equation. The velocity profiles were then normalized by the free stream velocity to produce plots of $\frac{u}{U_\infty}$ vs. y. the boundary layer thickness was determined by interpolating each profile to the location where $\frac{u}{U_\infty} = 0.99$.

These experimental methods also have limitations that affected the results. First, the pressure rake only measured velocity at specific locations perpendicular to the wall, so the exact 99% boundary layer thickness had to be interpolated rather than measured directly. This made the result vary heavily with the pressure probe spacing, especially close to the outer edge of the boundary layer. Additionally, the free stream velocity near the rake may not have been identical to the tunnel reference velocity, U_{ref}, especially close to the sting where the blockage and pressure gradient effects changed the flow. The power law approximation also required the boundary layer origin to be at the leading edge of the sting, but the actual starting point of the boundary layer was unknown because the flow had already traveled through the contraction section and trip strip. Finally, this experiment only measured boundary layer growth at two streamwise locations, so the flow development around the sting could not be entirely mapped.

The rest of this report is organized as follows. Section II describes the wind tunnel, sting configuration, pressure rake, instrumentation, and other hardware as well as the data acquisition procedure. Section III presents the measured velocity profiles, boundary layer thickness calculations, turbulent power law predictions, and discussion of sting effects. Section IV summarizes the main conclusions and limitations of the experiment.

II. Apparatus and Techniques

A. Wind Tunnel and Test Section

This experiment was performed in the Virginia Tech Stability Wind Tunnel. This tunnel runs under subsonic flow in a closed loop, single return system. The test section is approximately 7.32 m long and has a square cross section measuring 1.85 m by 1.85 m. The tunnel equipped with a variable speed motor and propeller system. This lets the velocity of the test section be adjusted for each required Reynolds number in the specific sweep. The wind tunnel can reach speeds of up to 75 m/s and a Reynolds numbers per meter of up to 5 million. Figure 1 shows the setup of the Virginia Tech Stability Wind Tunnel in greater detail.

For this experiment, the wind tunnel is characterized by a hard-wall configuration that was also anechoic. This means that it prevented sound waves from generating or traveling inside the test section of the tunnel. The hard-wall setup is made from aluminum panels that each measured 0.61 m by 0.61 m in the shape of a sqaure. These panels were numbered by their streamwise station, P1 to P9, which allowed the pressure rake location to be clearly identified before each test was run. Figure 2 displays a detailed view of each panel location, P1 to P9. After the flow passes through a settling chamber and contraction section to reduce turbulence, the flow enters the test section. A serrated trip strip is located upstream from the test section to enforce similar boundary layer development on all four walls.

The test section was completely empty except for the sting mount and other measurement tools. Nothing was mounted onto the sting for this experiment because the purpose of this experiment was to isolate the effect of the sting itself on the boundary layers that were measured on the walls. This was important because future testing may use the sting to mount a specific airfoil model or test body, so the disturbances caused by the sting itself must be fully understood beforehand.

B. Sting Geometry and Test Setup

Two different sting geometries were used in this experiment. The first geometry test was the baseline setup with no fillet added on the bottom of the sting. This no fillet case provided reference for how the sting mount affected the wall boundary layer without any additional shaping. The second sting geometry tested included a small fillet on the bottom of the sting. The purpose of the fillet was to reduce the separation close to the leading edge of the sting that would in turn weaken the horseshoe vortex formation. Figure 3 displays both geometric setups used.

The boundary layer pressure rake was set up at two different streamwise locations so that the boundary layer could be measured both downstream and upstream. The first upstream location was used as the less disturbed reference measurement, while the second downstream location was used to measure how the boundary layer grew.

The rake was mounted on the floor at panels 10 and 8, with panel 10 being closest to the leading edge of the sting and panel 8 being two panels downstream. The no fillet sting data was collected by this group, while the bottom fillet sting data was collected by the following group of students.

C. Boundary Layer Rake and Pressure Instrumentation

Boundary layer profiles were measured using a pneumatic pressure rake. The rake was mounted perpendicular to the wall so that the probes measured the total pressure at different distances from the wall. The rake was 154.2 mm tall and is shaped like a symmetric airfoil with a chord length of 62.5 mm. The airfoil shape reduced blockage effects. The rake body had a chord of 42.5 mm and the probe tips extended 20 mm past the main body. Figure 4 shows a detailed view of the pressure rake and its main body.

The rake contained a total of 30 stagnation pressure probes each arranged and distanced with semilogarithmic spacing. The first 11 probes were evenly spaced over the first 10.5 mm from the wall where the frictional effects on the velocity gradient were expected to be the greatest. The rest of the following probes were spaced logarithmically farther from the wall that reached a maximum height of 152.6 mm. This specific spacing allowed the pressure rake to capture both the effects close to the wall and the outer area of the velocity profile.

The rake was manufactured with stainless steel and included internal wiring that connected each probe to pressure tubing. The rake was bolted to a 0.61 m by 0.61 aluminum floor panel using four countersunk screws. The panel that held the pressure rake also had two static pressure ports, which were used to measure the static pressure directly at the rake location. The wiring connecting the 30 rake probes to the static pressure ports was connected to a DTC Initium ESP32HD pressure scanner. This pressure scanner had a range of plus or minus 20 inWC which is plus or minus 0.05% in terms of accuracy. The pressure scanner is shown in Figure 5.

D. Tunnel Data Acquisition

The reference velocity in the test section, U_{ref}, was measured using the pressure difference between the static ports in the settling chamber and the contraction section. The test section system also recorded the test section temperature and ambient pressure. The flow temperature was measured with an Omega thermistor, and the ambient pressure was measured with an Omega MMB26 pressure transducer. With these measurements, air density and viscosity could be calculated via the ideal gas law and Sutherlands relations. These properties all affect the Reynolds number and velocity calculations.

During each test run, the data acquisition system recorded the rake total pressures, static pressures at that location, the reference velocity, flow temperature, and the ambient pressure. The MATLAB acquisition program saved the pressure data during each Reynolds number condition. For each speed, 25 records were taken to ensure a reliable average pressure value. After each sweep, the data files were processed into MATLAB for analysis.

Bernoulli's equation in the form of equation (6) was used to obtain a velocity profile. The velocity profiles were then normalized by the free stream velocity to produce plots of $\frac{u}{U_\infty}$ vs. y for further data analysis.

E. Test Procedure and Conditions

At each pressure rake location, boundary layer profiles were measured over a sweep of Reynolds numbers. Each Reynolds number per meter measured were targeted at approximately: 1.15 million per meter, 1.7 million per meter, 2.25 million per meter, 2.8 million per meter, 3.35 million per meter, and 3.9 million per meter. Although, the actual values measured are slightly different. The tunnel speed was adjusted at each condition so that the target Reynolds number was reached and held constant. Each Reynolds number was kept within 0.05 million per meter of the target.

For each Reynolds number, the pressure rake measured the stagnation pressure distribution across the boundary layer. After the data was collected at the first streamwise location, the rake was then moved and set up at the second streamwise location, and the Reynolds number sweep was then repeated. This allowed boundary layer growth to compare between two different streamwise locations. The same process was performed for the no fillet and fillet geometries, which also allowed for the effects of the fillet to be shown in the data.

In this experiment, the rake was mounted on the floor of the test section. The two rake stations were located at panels 10 and 8. The Reynolds number range used in the final analysis was approximately 1.15 million to 3.9 million per meter, which corresponds to a velocity of about 20.3 to 71.5 m/s.

F. Experimental Limitations

There were several limitations to the test setup that affected the accuracy of the measurements. First, the rake only performed pressure measurements at 30 specific locations perpendicular to the wall, so the boundary layer thickness was not measured directly, it was only estimated. The point where $\frac{u}{U_\infty} = 0.99$ had to be interpolated between measured probe location. This made the calculated thickness extremely vary heavily with the spacing of each probe, especially close to the outer edge of the boundary layer.

Secondly, the rake itself had a specific size and although it was shaped as a symmetric airfoil, it could still slightly disturb the flow being measured. The rake body still took up space within the test section, therefore the blockage effects are technically not negligible. Additionally, the freestream velocity near the rake was not always the same as the reference velocity, more importantly near the sting where the blockage and pressure gradients were the greatest. Finally, only two streamwise locations were measured in this experiment, so the full growth and development of the boundary layer could not be entirely mapped and shown.

III. Results and Discussion

A. Boundary Layer Velocity Profiles

Boundary layer velocity profiles were graphed for both rake locations on both panels 8 and 10, as well as for both sting geometries. The profiles were graphed as $\frac{u}{U_\infty}$ vs. the position *y* that is perpendicular to the wall, where *u* is the streamwise velocity measured by the pressure rake, and U_∞ is the free stream velocity outside of the boundary layer. The boundary layer thickness δ was determined by interpolating each velocity profile where $\frac{u}{U_\infty} = 0.99$.

Figures 6 and 7 show the no fillet velocity profiles at panels 8 and 10. Figures 8 and 9 show the fillet velocity profiles also at panels 8 and 10. At both positions, the velocity profiles showed the expected turbulent boundary layer shape. Near the wall, the velocity was lowest because of the no slip condition and frictional effects. As the distance perpendicular to the wall increased, the velocity increased as well until it reached the velocity free stream value. The plots were not linear, as expected by boundary layer theory.

For both sting geometries, increasing the Reynolds number made the boundary layer thinner most of the time. This is because the higher Reynolds number velocity profiles reached $\frac{u}{U_\infty} = 0.99$ at a smaller *y* distance. This is consistent with the expected boundary layer trend, where the boundary layer thickness decreases as the Reynolds number increases at a fixed location. However, the $Re = 3.9$ million per meter test run did not remain as consistent to this trend. For both the sting geometries, the boundary layer thickness increased again at $Re = 3.9$ million per meter. This suggests that the Reynolds number data may have been affected by inconsistencies during testing such as uncertainties in choosing the freestream velocities.

B. Difference between U_∞ and U_{ref}

The free stream velocity U_∞ and the tunnel reference velocity U_{ref} are not the same quantity. The reference velocity is the tunnel speed measured by the wind tunnel measurement system and represents the reference flow condition used to set the Reynolds number. The free stream velocity is the velocity directly outside the boundary layer at the pressure rake location.

Understanding the difference between these two quantities is important because the sting mount can change the flow near the wall. Since the sting takes up space within the tunnel, it can create blockage and pressure gradient effects. These effects can make the flow velocity right outside the boundary layer different from the tunnel reference velocity. Therefore, U_{ref} was more appropriate to use for setting the Reynolds number for each test run, but U_∞ was better used for normalizing the velocity profiles to determine the boundary layer thicknesses.

C. Boundary Layer Thickness Results

The measured boundary layer thicknesses for the no fillet and fillet geometries are shown in Table 1. panel 8 was treated as the upstream location, and panel 10 was treated as the downstream location.

Table 1. Measured boundary layer thicknesses for the no fillet and fillet geometries at panels 8 and 10.

Re (million per meter)	No fillet δ, Panel 8 (m)	No fillet δ, Panel 10 (m)	Fillet δ, Panel 8 (m)	Fillet δ, Panel 10 (m)
1.15	0.103969	0.123729	0.110275	0.120270
1.70	0.093335	0.109134	0.096332	0.110850
2.25	0.090360	0.102656	0.092511	0.106487
2.80	0.086014	0.099061	0.085739	0.100592
3.35	0.084889	0.100160	0.085468	0.099740
3.90	0.099514	0.113790	0.099418	0.111810

For both sting geometries, the boundary layer thicknesses were larger at panel 10 than at panel 8. This shows that the boundary layer grew as it moved between the two pressure rake positions. This is consistent with the expected behavior of a turbulent boundary layer as it should grow as the flow travels further downstream.

For the geometry with no fillet, the boundary layer thickness at panel 8 decreased from 0.103969 meters at $Re = 1.15$ million per meter to 0.085468 meters at $Re = 3.35$ million per meter, before increasing back to 0.099481 meters at $Re = 3.90$ million per meter. At panel 10, the thickness decreased from 0.120270 meters to 0.099740 meters, before increasing back to 0.11810 meters at the highest Reynolds number.

Overall, the lowest Reynolds number produced the thickest boundary layer, while the higher Reynolds number generally produced a thinner boundary layer except for $Re = 3.90$ million per meter.

D. Streamwise Boundary Layer Growth Rate

The streamwise growth rate of the boundary layer was estimated using the following equation:

$$\frac{d\delta}{dx} = \frac{\delta_{P10} - \delta_{P8}}{\Delta x} \tag{7}$$

where $\Delta x = 1.22$ meters, which was the distance between panels 8 and 10. Table 2 displays the calculated boundary layer growth rate for both sting geometries.

Table 2. Calculated boundary layer growth rates for the no fillet and fillet geometries between panels 8 and 10.

Re (million per meter)	No fillet growth rate	Fillet growth rate
1.15	0.01620	0.00819
1.70	0.01295	0.01190
2.25	0.01008	0.01146
2.80	0.01069	0.01218
3.35	0.01252	0.01170
3.90	0.01170	0.01016

The geometry with no fillet had an average growth rate of approximately 0.01236 while the geometry with a fillet had an average growth rate of approximately 0.01093. This means that the filleted geometry had a lower average growth rate between panels 8 and 10 compared to the no fillet sting. This suggests that the fillet reduced the streamwise growth rate of the boundary layer by reducing the strength of the disturbances seen at the leading edge of the sting.

This trend, however, was not seen at every Reynolds number. At $Re = 2.25$ and at $Re = 2.80$ million per meter, the fillet growth rate was slightly higher than the no fillet growth rate. Because these differences were very small, the fillet effect still reduced the growth rate on average overall.

E. The Effect of Fillet Geometry

The fillet geometry was expected to reduce the flow disturbances, such as the horseshoe vortices, caused by the sting because the smooth fillet decreases the flow separation seen by the sting body. At the upstream panel location, panel 8, the fillet produced slightly thicker boundary layers than the no fillet geometry. For example, at $Re = 1.15$ million per meter, the filleted boundary layer thickness was 0.110275 meters, while the no fillet case saw a thickness of 0.103969 meters. Although, at higher Reynolds numbers, the upstream thickness values were essentially the same.

At the downstream location, panel 10, the effect of the fillet was mixed. The fillet geometry produced a thinner boundary layer than the no fillet case at $Re = 1.15$, $Re = 3.35$, and $Re = 3.90$ million per meter. However, the fillet produced a thicker boundary layer at $Re = 1.70$, $Re = 2.25$, and $Re = 2.80$ million per meter. Table 3 below displays the percent difference of boundary layer thicknesses that the filleted geometry creates.

Table 3. Calculated percent difference in boundary layer thicknesses between the two sting geometries at panel 10.

Re (million per meter)	Fillet δ_{P10} – No Fillet δ_{P10} (m)	Percent Difference
1.15	-0.003459	-2.80%
1.70	0.001716	1.57%
2.25	0.003831	3.73%
2.80	0.001513	1.55%
3.35	-0.000420	-0.42%
3.90	-0.001980	-1.74%

The downstream boundary layer thicknesses for both geometries were close overall. The average downstream thickness for the no fillet geometry was 0.10809 meters, while the average downstream thickness for the fillet geometry was 0.10829 meters. This is only a 0.19% percent difference, meaning the fillet did not consistently produce a thinner boundary layer all the time.

However, the boundary growth between panels 8 and 10 was lower on average for the fillet geometry. This suggests that the fillet reduced the extra boundary layer growth caused by the sting between the two panels.

F. Comparison with the Turbulent Power Law Prediction

The measured boundary layer thickness can be compared with the turbulent power law approximation that is given by equation (3). This equation is an approximation for a turbulent boundary layer on a flat plate, it is not an exact solution. It assumes that there are no pressure gradients or blockage effects, therefore this approximation is expected to not exactly match the results of this experiment.

In order to calibrate the exact x coordinate where the boundary layer begins, Panel 10 can be used as it is first to encounter the flow. Using equation (3) for each Reynolds number and corresponding boundary layer thicknesses, the effective x location was solved for. The $Re = 3.90$ million per meter test run was not included in the calibration calculation since it didn't not follow the decreasing boundary layer thickness trend. Using and averaging the remaining five Reynolds number test runs, the calibrated panel 10 location was approximately:

$$x_{P10} = 8.04 \text{ m} \tag{8}$$

for the non fillet geometry test runs, and

$$x_{P10} = 8.11 \text{ m} \tag{9}$$

for the fillet geometry test runs. Since panel 8 was positioned 1.22 meters farther downstream from panel 10, the calibrated locations were:

$$x_{P8} = 8.04 + 1.22 = 9.26 \text{ m} \tag{10}$$

for the no fillet test runs and

$$x_{P8} = 8.11 + 1.22 = 9.26 \text{ m} \tag{11}$$

for the fillet test runs. All of these calibrated x values were then used in the turbulent power law equation, equation (3), to predict what the boundary layer thickness is expected to be at panel 8. This is reflected in Table 4 below:

Table 4. Measured and predicted boundary layer thicknesses at panel 8 using the calibrated power law model.

Re (million per meter)	No fillet measured δ_{P8} (m)	No fillet expected δ_{P8} (m)	Fillet measured δ_{P8} (m)	Fillet expected δ_{P8} (m)
1.15	0.103969	0.134711	0.110275	0.135498
1.70	0.093335	0.124581	0.096332	0.125310
2.25	0.090360	0.117789	0.092511	0.118478
2.80	0.086014	0.112748	0.085739	0.113408
3.35	0.084889	0.108776	0.085468	0.109412
3.90	0.099514	0.105518	0.099418	0.106136

For both of the sting geometries, the calibrated power law equation predicted that the boundary layers at panel 8 should be thicker than the measured value. This is because the power law assumes that the boundary layer continues to grow downstream like a flat plate turbulent boundary layer. However, the measured boundary layer thicknesses were smaller for every Reynolds number.

This difference shows specifically that the flow between panels 10 and 8 did not behave like an ideal smooth flat plate turbulent boundary layer. A likely explanation is that the sting geometry changed the nearby pressure field surrounding the area. Since panel 10 was located near the sting, the boundary layer at that position may have already been affected by the adverse pressure gradient or blockage effects. When these effects were taken into account and were used to calibrate panel 10, the equation used still assumed that the flow between the panels would still cause the expected flat plate boundary layer growth. Instead, the measured panel 8 boundary layer thicknesses were thinner than expected. This means that effects from the sting, such as pressure changes that in turn changed velocities, or vortices likely changed the boundary layer in ways that the power law could not predict.

IV. Conclusion

An experiment was performed to investigate the turbulent boundary layer growth on the floor and ceiling of the Virginia Tech Stability Wind Tunnel with an added sting mount in the test section. Boundary layer profiles were measured using a pneumatic pressure rake at two different positions: panel 10 which was right near the sting, and panel 8 that was farther downstream. Two different sting geometries were also compared: one with an added fillet on the bottom, and one without. Pressure measurements from the rake were transformed into velocity profiles via Bernoulli's equation. Boundary layer thicknesses were determined at the location where $\frac{u}{U_\infty} = 0.99$. The results that were measured were then compared with the turbulent flat plate power law predictions. The following conclusions can be drawn:

1. The measured velocity profiles had an expected velocity profile shape. The velocity was lowest near the wall and increased as the distance away from wall increased until it reached the free stream velocity value.
2. For both of the sting geometries, the boundary layer thickness mostly decreased as Reynolds numbers increased from $Re = 1.15$ million per meter to $Re = 3.35$ million per meter. This is consistent

with the expected turbulent boundary layer relationship that higher Reynolds numbers will produce thinner boundary layers at fixed locations.

3. The $Re = 3.90$ million per meter test runs did not follow the same trend as clearly because the boundary layer thickness increased. This suggests that this condition may have been affected by uncertainties or additional flow disturbances.
4. Comparing the two sting geometries showed that the added fillet did not consistently produce a smaller boundary layer. However, on average the fillet test runs did have a lower streamwise rate of growth between panels 8 and 10, This suggests that the fillet did reduce some of the flow disturbances caused by the presence of the sting in the test section.
5. The power law relation was used as a baseline comparison against the measured results, and they did not match. After calibrating the power law using panel 10, the predicted panel 8 boundary layer thicknesses were larger than the measured values. This means that the flow did not behave like an ideal flat plate.
6. The main sources of inconsistencies between the experiment and the power law prediction were the pressure gradients and blockage effects induced by the sting. Additional sources of disagreement could be possible horseshoe vortex motion and uncertainty in the effective boundary layer starting point.

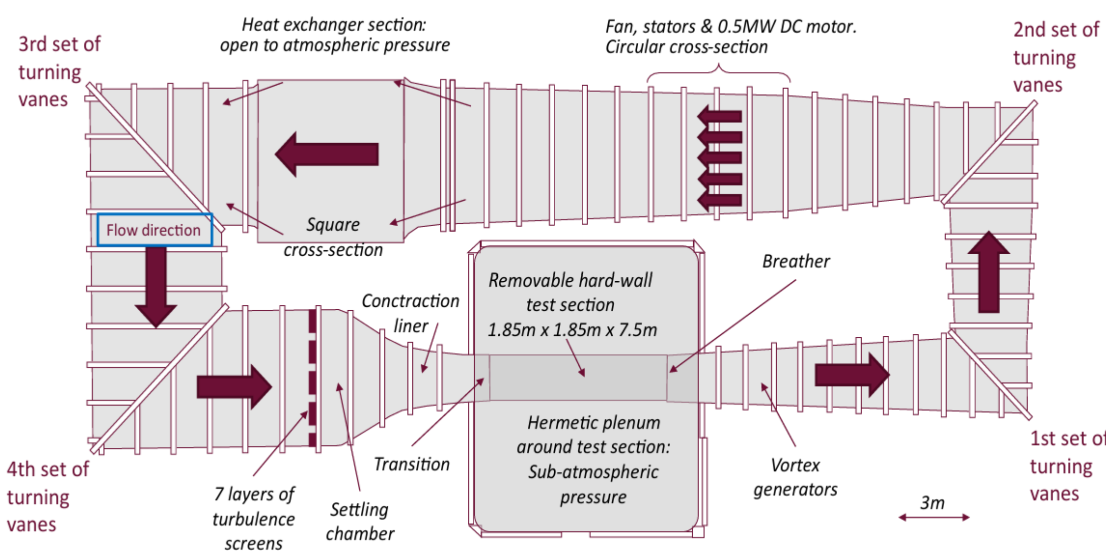

Fig. 1. Diagram of Virginia Tech Stability Wind Tunnel configuration, adapted from Borgoltz and Intaratep (2026).

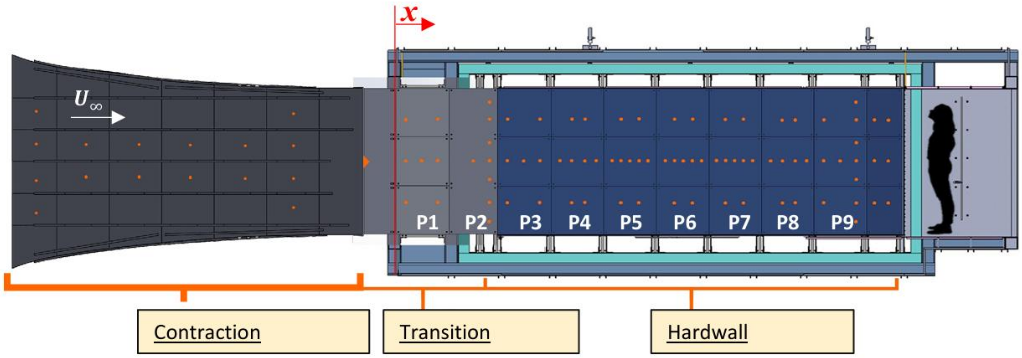

Fig. 2. Diagram depicting a side view of the Virginia Tech Stability Wind Tunnel's contraction and test section in the hard wall configuration with panels and pressure tap locations shown, adapted from Borgoltz and Intaratep (2026).

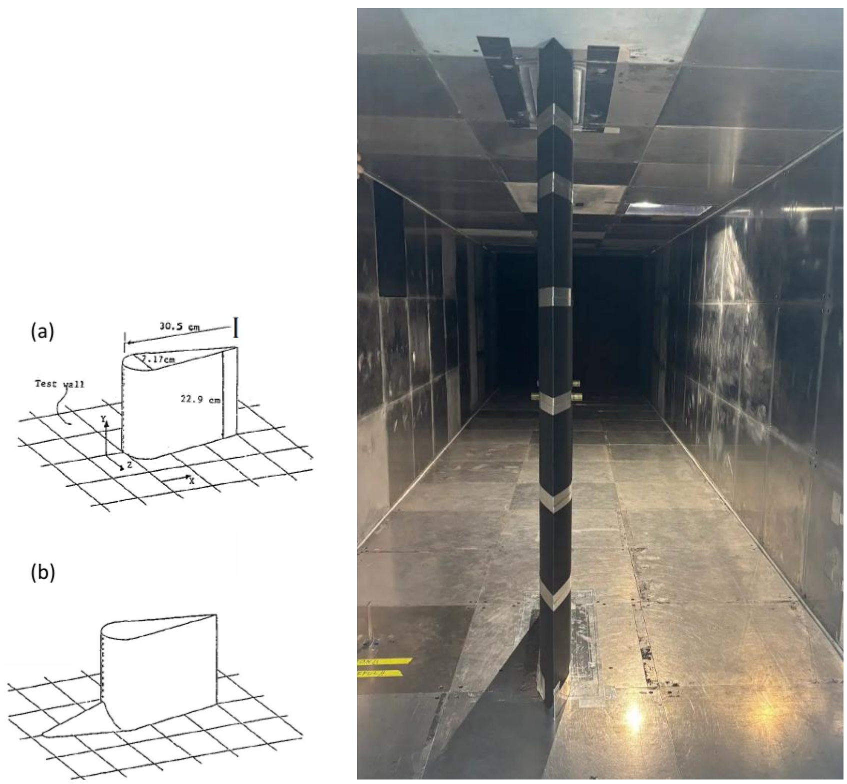

Fig. 3. Diagram and pictures of each sting geometry. Left side (a) shows the sting dimensions without the fillet applied, while (b) shows the sting geometry with the fillet applied, both adapted from W.J. Devenport. The picture on the right side shows the sting to scale inside the tunnel.

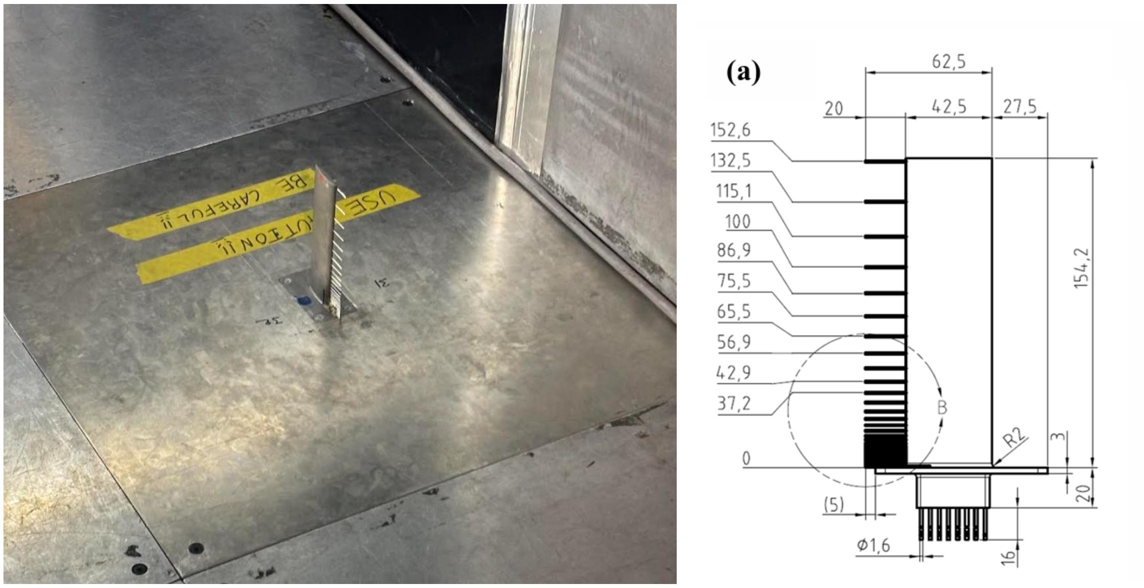

Fig. 4. Left side picture shows the pneumatic pressure rake used to measure the boundary layer velocity profile. The right image (a), adapted from Borgoltz and Intaratep (2026), shows the rake dimensions and probe spacing.

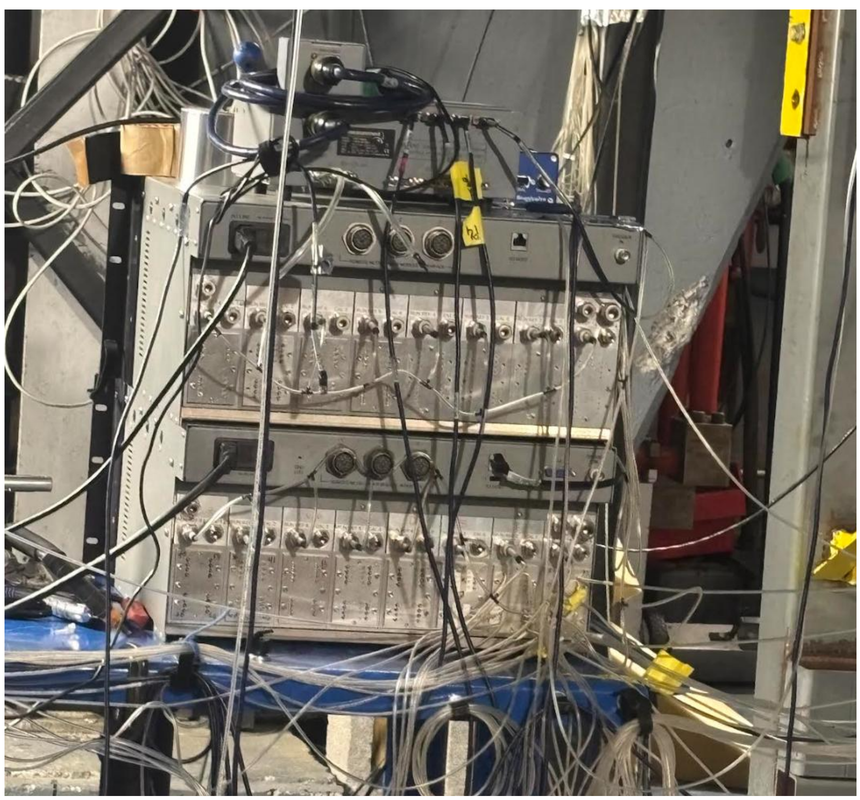

Fig. 5. DTC Initium ESP32HD pressure scanner that connected the 30 pressure rake probes to the static pressure ports. This pressure scanner has a range of plus or minus 20 inWC which is plus or minus 0.05% in terms of accuracy.

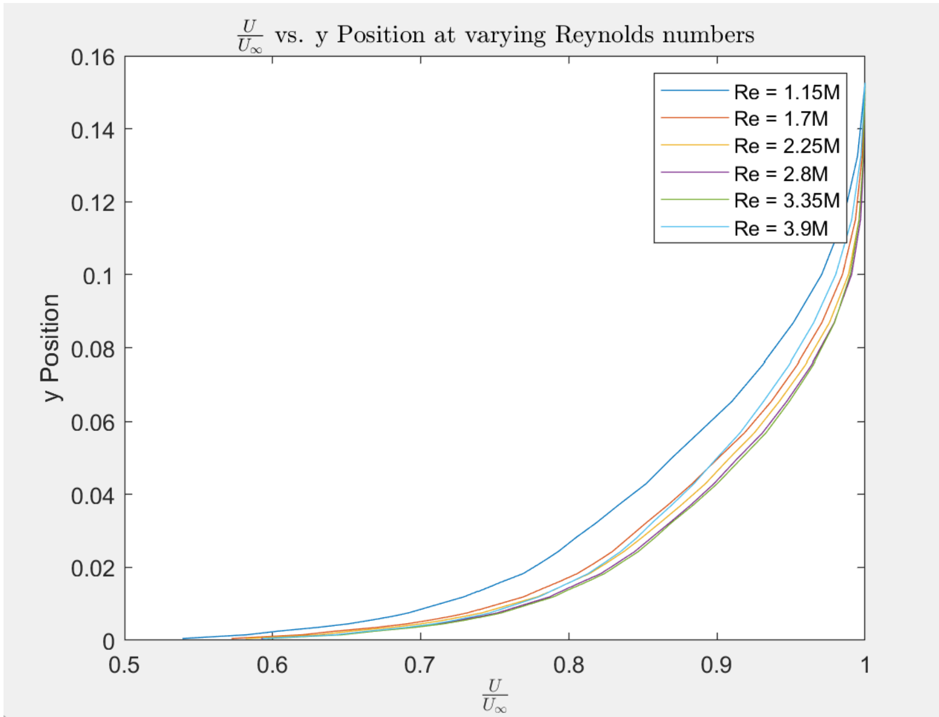

Fig. 6. No fillet boundary layer profiles at Panel 10 for Reynolds numbers 1.15 million per meter to 3.9 million per meter.

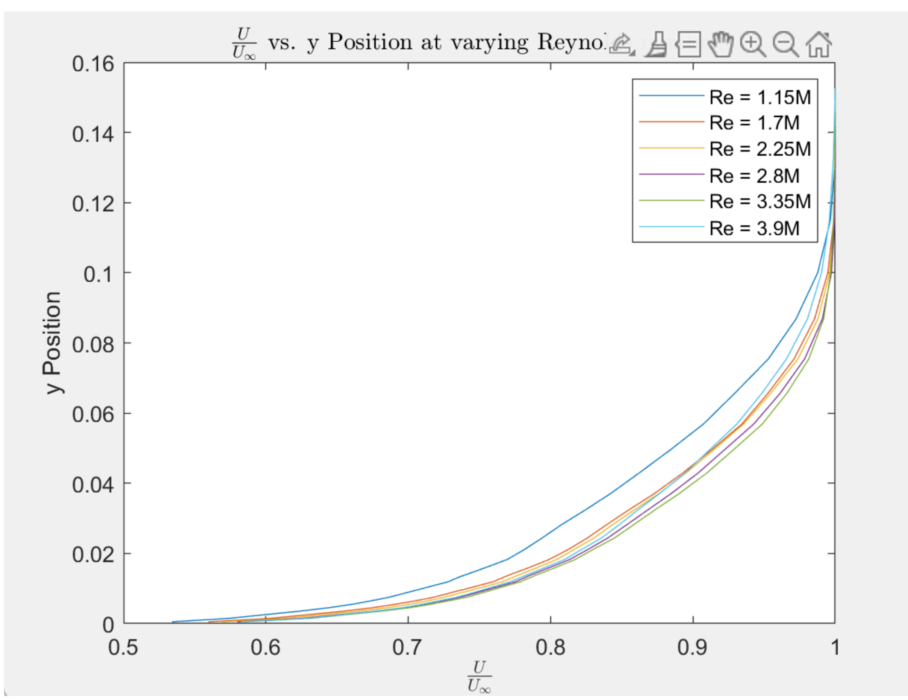

Fig. 7. No fillet boundary layer profiles at Panel 8 for Reynolds numbers 1.15 million per meter to 3.9 million per meter.

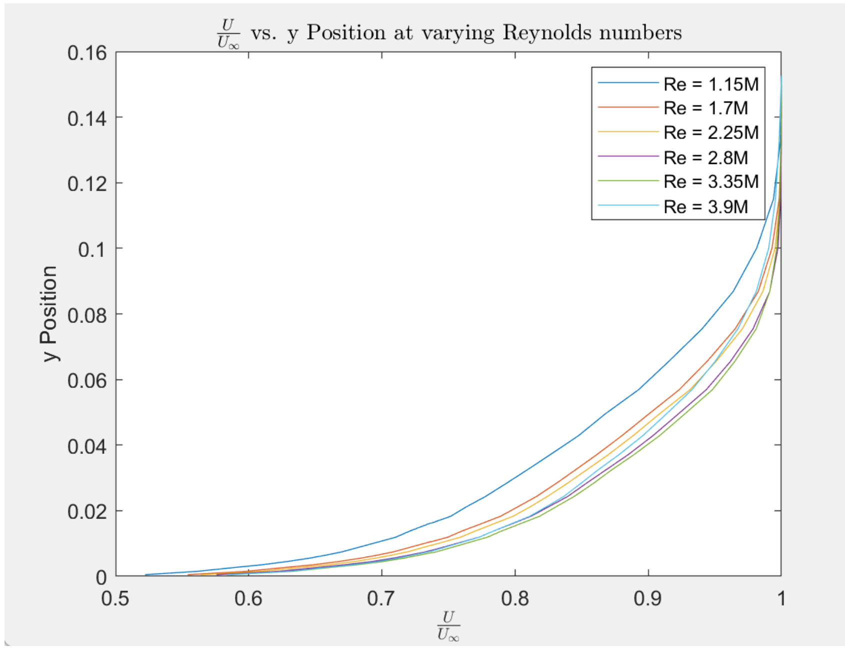

Fig. 8. Fillet boundary layer profiles at Panel 8 for Reynolds numbers 1.15 million per meter to 3.9 million per meter.

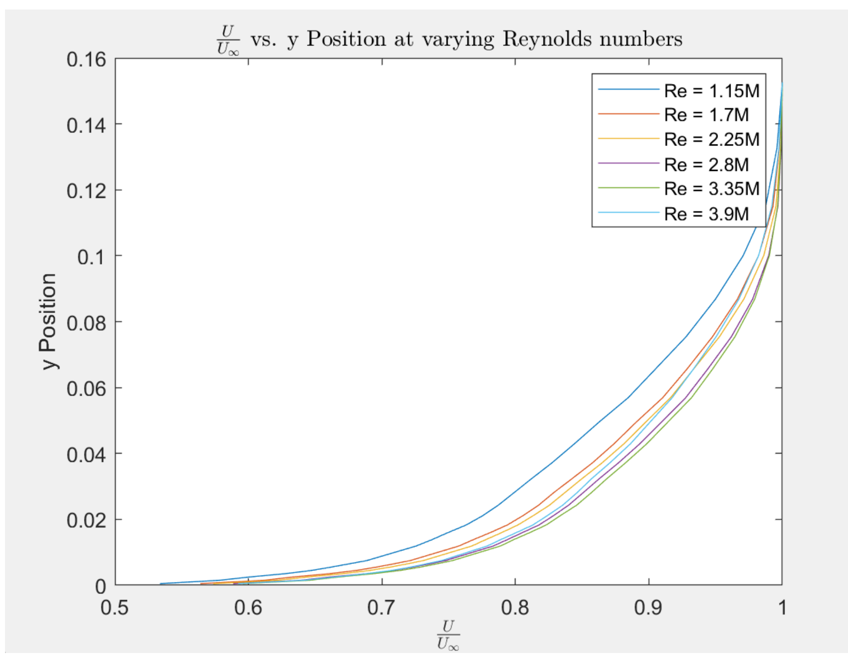

Fig. 9. Fillet boundary layer profiles at Panel 10 for Reynolds numbers 1.15 million per meter to 3.9 million per meter.

REFERENCES

Borgoltz, A., Intaratep, N., and Szőke, M., 2026, *AOE 3054 Experimental Methods Course Manual. Boundary Layer Growth on Wind Tunnel Walls in the Presence of a Sting Mount, Kevin T. Crofton Department of Aerospace and Ocean Engineering, Virginia Tech, Blacksburg, VA.*

Appendix

A. Uncertainty Analysis

Uncertainty was calculated for measured quantities in order to estimate the total uncertainty in the boundary layer thicknesses. The primary sources of uncertainty came from atmospheric pressure, stagnation pressure, flow temperature, free stream velocity, probe locations, and the interpolation used to determine the boundary layer thicknesses. Since the local velocities, normalized velocities, and boundary layer thicknesses were determined indirectly from these values, uncertainties must be taken into account.

Uncertainty in any derived results R calculated from measured variables *a,b,c* was determined using the root-sum-square equation which is shown below:

$$\delta(R) = \sqrt{\left(\frac{\partial R}{\partial a}\delta(a)\right)^2 + \left(\frac{\partial R}{\partial b}\delta(b)\right)^2 + \left(\frac{\partial R}{\partial c}\delta(c)\right)^2 + \cdots} \quad (12)$$

The partial derivative terms were calculated using a spreadsheet perturbation method. Each primary measurement was perturbed by its uncertainty one at a time while the other values were held constant. This method was used and is reflected in tables 5 to 8 below:

Table 5. Table for calculation of uncertainty in local probe velocity.

			Perturbation			
	Quantity	**Uncertainty**	*a+da,b,c,d*	*a,b+db,c,d*	*a,b,c+dc,d*	*a,b,c,dd*
Atmospheric pressure (Pa)	101325	10.2	101335.2	101325	101325	
Local Static Pressure (Pa)	24884	2.49	24884	24886.49	24884	24884
Local Stagnation Pressure (Pa)	49768	2.49	49768	49768	49770.49	49768
Flow Temperature (K)	300.15	0.20	300.15	300.15	300.15	300.35
Intermediate results						
Density (kg/m^3)	1.176036		1.176036	1.176036	1.176036	
Final Result						
Local speed (m/s)	205.7144		205.7041	205.7041	205.7247	205.7829
		Change	+0.010353	+0.010286	-0.01029	-0.06853
Uncertainty in final result (m/s)	0.070814					

Table 6. Table for calculation of uncertainty in normalized local probe velocity.

	Quantity	Uncertainty	Perturbation *a+da,b,c,*	*a,b+db,c,d*	*a,b,c+dc,d*	*a,b,c,dd*
Atmospheric pressure (Pa)	101325	10.2	101335.2	101325	101325	
Local Static Pressure (Pa)	24884	2.49	24884	24886.49	24884	24884
Local Stagnation Pressure (Pa)	49768	2.49	49768	49768	49770.49	49768
Flow Temperature (K)	300.15	0.20	300.15	300.15	300.15	300.35
Freestream Velocity (m/s)	300	0.07	300	300	300	300.07
Intermediate results						
Density (kg/m^3)	1.176036		1.176036	1.176036	1.176036	
Local speed (m/s)	205.7144		205.7144	205.7144	205.7144	
Final Result						
Normalized speed (m/s)	0.685715		0.685658	0.685749	0.685749	0.685553
		Change	+0.00003	+0.00003	-0.00023	+0.000162
Uncertainty in final result (m/s)	0.000236					

Table 7. Table for calculation of uncertainty in boundary layer thickness.

	Quantity	Uncertainty	Perturbation *a+da,b,c,d*	*a,b+db,c,d*	*a,b,c+dc,d*	*a,b,c,dd*
Vertical location of measurement 1 (m)	9	0.01	9.01	9	9	
Vertical location of measurement 2 (m)	10	0.01	10	10.01	10	10
Normalized Velocity 1 (m/s)	0.7	0.01	0.7	0.7	0.77	0.7
Normalized Velocity 2 (m/s)	0.95	0.01	0.95	0.95	0.95	0.95
Final Result						
Boundary Layer Thickness (m)	10.16		10.1584	10.1716	10.22323	9.903952
		Change	+0.0016	-0.0116	-0.06323	+0.256048
Uncertainty in final result (m)	0.264					